Thread

Experiment

Experiment #1

แก้ไข for ทั้ง 2 จุด

```
for(i=0; i < 100000; i++)
```

```
1  // simple thread
2  using System;
3  using System.Threading;
4
5  namespace Lab_OS_Concurrency
6  {
7      class Program
8      {
9          static void TestThread1()
10         {
11             int i;
12             for (i = 0; i < 100; i++)
13                 Console.WriteLine("Thread# 1 i={0}", i);
14         }
15         static void TestThread2()
16         {
17             int i;
18             for (i = 0; i < 100; i++)
19                 Console.WriteLine("Thread# 2 i={0}", i);
20         }
21
22         static void Main(string[] args)
23         {
24             Thread th1 = new Thread(TestThread1);
25             Thread th2 = new Thread(TestThread2);
26             th1.Start();
27             th2.Start();
28         }
29     }
30 }
```

Experiment #2

- Resource sharing among threads

```
1   //test resource sharing
2   using System;
3   using System.Threading;
4
5   namespace Lab_OS_Concurrency01
6   {
7       class Program
8       {
9           static int resource = 10000;
10          static void TestThread1()
11          {
12              Console.WriteLine("Thread# 1 i={0}", resource);
13          }
14          static void TestThread2()
15          {
16              Console.WriteLine("Thread# 2 i={0}", resource);
17          }
18
19          static void Main(string[] args)
20          {
21              Thread th1 = new Thread(TestThread1);
22              Thread th2 = new Thread(TestThread2);
23              th1.Start();
24              th2.Start();
25          }
26      }
27  }
```

Experiment #3

- Pause a thread

แก้ไข

`Thread.Sleep(100)`

```
//test pause a thread
using System;
using System.Threading;

namespace Lab_OS_Concurrency02
{
    class Program
    {
        static int resource = 10000;
        static void TestThread1()
        {
            resource = 55555;
        }

        static void Main(string[] args)
        {
            Thread th1 = new Thread(TestThread1);
            th1.Start();
              Thread.Sleep(10);
            Console.WriteLine("resource={0}", resource);
        }
    }
}
```

Experiment #3.1

1. Adjust the sleep time so that the output is printed immediately after the last "." is displayed, as shown in the example on the next page.
2. Suppose this program, using the sleep time you found in Step 1, is run on another computer that is either faster or slower than the one used in this experiment. Will the output still be the same? Explain your answer.

```
1   //test pause #2
2   using System;
3   using System.Threading;
4
5   namespace Lab_OS_Concurrency01
6   {
7       class Program
8       {
9           static int resource = 10000;
10          static void TestThread1()
11          {
12              int i;
13              for (i = 0; i < 45555; i++)
14              {
15                  resource++;
16                  Console.Write(".");
17              }
18          }
19
20          static void Main(string[] args)
21          {
22              Thread th1 = new Thread(TestThread1);
23              th1.Start();
24              Thread.Sleep(10);
25              Console.WriteLine("Resource = {0}", resource);
26          }
27      }
28  }
```

Experiment #3.1 desired result



Experiment #3.2

- Join thread

```
1   //test pause #2
2   using System;
3   using System.Threading;
4
5   namespace Lab_OS_Concurrency01
6   {
7       class Program
8       {
9           static int resource = 10000;
10          static void TestThread1()
11          {
12              int i;
13              for (i = 0; i < 45555; i++)
14              {
15                  resource++;
16                  Console.Write(".");
17              }
18          }
19
20          static void Main(string[] args)
21          {
22              Thread th1 = new Thread(TestThread1);
23              th1.Start();
24              //Thread.Sleep(10);
25              th1.Join();
26              Console.WriteLine("Resource = {0}", resource);
27          }
28      }
29  }
```